

VALENTINE, NE, USA

WMO: 725670

Lat: 42.857N

Lon: 100.548W

Elev: 789

StdP: 92.19

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 24032

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.9	-19.6	-26.6	0.4	-22.5	-23.4	0.5	-19.0	14.3	-4.1	12.4	-2.8	3.4	280	0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	36.6	20.7	34.5	20.5	32.5	20.2	23.3	31.1	22.4	30.4	21.6	29.5	6.3	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.1	17.4	26.6	20.1	16.3	26.0	19.1	15.3	25.3	73.4	31.2	69.9	30.3	66.6	29.4	26.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12.5	11.1	9.8	-28.9	39.8	3.2	2.0	-31.2	41.3	-33.1	42.5	-34.9	43.6	-37.3	45.1
			-29.0	24.8	3.0	0.8	-31.1	25.3	-32.9	25.8	-34.6	26.2	-36.8	26.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	9.3	-4.4	-3.0	2.7	8.1	14.1	20.1	24.1	22.8	17.6	9.4	2.3	-3.7
	DBStd	11.57	7.13	7.29	7.06	5.81	4.98	4.21	3.64	3.47	5.14	5.49	6.30	6.65
	HDD10.0	1911	445	364	239	101	18	1	0	0	4	77	238	424
	HDD18.3	3798	703	596	486	309	149	28	3	6	74	279	481	683
	CDD10.0	1638	0	1	12	43	144	304	437	397	233	59	7	0
	CDD18.3	484	0	0	0	2	17	82	182	146	53	3	0	0
	CDH23.3	6242	0	0	17	72	303	1034	2253	1674	777	104	8	0
	CDH26.7	2901	0	0	2	19	103	432	1169	801	344	31	1	0
Wind	WSAvg	4.6	4.4	4.5	4.8	5.3	5.1	4.6	4.2	4.3	4.5	4.7	4.5	4.3
Precipitation	PrecAvg	476	7	12	25	48	79	79	77	56	38	25	16	8
	PrecMax	830	21	36	107	139	228	180	228	170	150	90	66	46
	PrecMin	268	0	2	0	6	11	11	7	3	8	0	0	0
	PrecStd	121	6	10	20	31	43	40	44	34	28	20	15	8
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	15.9	19.9	25.9	29.5	33.1	37.0	40.0	37.9	36.3	30.3	24.4	16.1
		MCWB	6.8	8.7	11.5	13.6	18.2	20.2	20.7	20.6	18.6	15.3	11.0	7.2
	2%	DB	11.5	14.6	21.4	25.2	29.5	33.4	37.4	35.6	33.5	26.2	19.5	11.7
		MCWB	5.0	6.5	10.3	12.9	17.0	20.6	20.8	20.6	18.9	13.9	9.3	5.1
	5%	DB	8.3	10.7	17.6	22.0	26.7	31.2	35.0	33.4	30.7	22.9	15.9	8.4
		MCWB	3.2	4.4	9.0	11.7	16.2	19.8	21.2	20.4	18.1	13.0	7.7	3.3
	10%	DB	5.5	7.4	14.2	18.8	24.0	29.0	32.9	31.2	28.1	19.7	12.3	5.3
		MCWB	1.8	3.1	7.0	10.2	14.8	19.0	21.0	20.0	17.2	11.5	6.1	1.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.3	9.2	13.4	15.4	20.1	23.6	24.6	24.0	22.0	17.5	11.9	7.8
		MCDB	15.0	18.2	20.8	24.7	28.8	31.1	32.5	32.3	29.7	24.5	21.3	15.2
	2%	WB	5.2	6.9	11.1	13.7	18.6	22.0	23.7	22.8	20.5	15.5	9.9	5.3
		MCDB	11.2	13.6	19.9	23.3	26.8	29.8	31.5	30.5	29.0	23.5	18.4	11.0
	5%	WB	3.5	5.0	9.3	12.2	17.1	21.0	22.8	21.9	19.5	13.7	8.0	3.5
		MCDB	7.9	10.5	17.1	20.8	25.0	28.9	30.8	29.5	27.7	21.1	15.6	8.1
	10%	WB	1.9	3.1	7.4	10.7	15.6	20.0	22.0	21.1	18.3	11.9	6.2	1.7
		MCDB	5.1	7.3	14.0	17.6	22.5	27.5	30.2	28.4	25.8	18.8	11.9	5.1
Mean Daily Temperature Range	5% DB	MDBR	13.2	13.1	14.0	13.8	13.6	14.0	14.7	14.8	15.6	14.9	14.5	13.0
		MCDBR	17.2	18.2	20.8	20.2	18.2	17.1	17.9	18.0	19.2	21.0	20.5	17.3
		MCWBR	11.3	10.9	11.2	9.9	8.2	6.8	5.9	6.3	7.4	10.2	11.3	11.3
	5% WB	MCDBR	16.2	17.3	19.1	18.1	15.6	14.8	15.2	15.0	16.0	18.2	19.4	16.6
		MCWBR	10.8	10.7	10.6	9.3	7.5	6.7	6.4	6.1	7.1	9.4	11.0	11.1
Clear-Sky Solar Irradiance	taub		0.259	0.268	0.291	0.328	0.350	0.362	0.383	0.381	0.341	0.307	0.271	0.261
	taud		2.534	2.514	2.494	2.393	2.417	2.413	2.391	2.374	2.456	2.521	2.543	2.512
	Ebn at Noon		905	949	960	940	922	908	884	875	890	887	880	866
	Edh at Noon		67	81	94	113	114	115	116	114	98	80	66	63
All-Sky Solar Radiation	RadAvg		2.09	2.97	4.05	5.06	6.03	6.86	7.00	5.93	4.80	3.27	2.26	1.74
	RadStd		0.15	0.18	0.33	0.31	0.50	0.46	0.24	0.33	0.29	0.34	0.16	0.13

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (13 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+76	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page